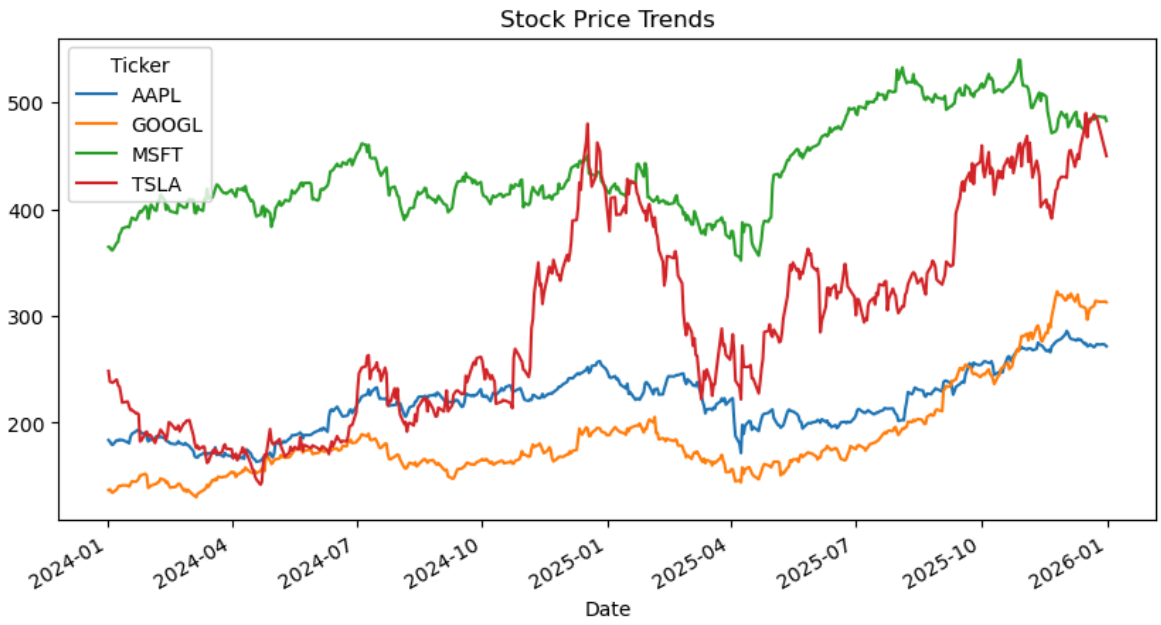
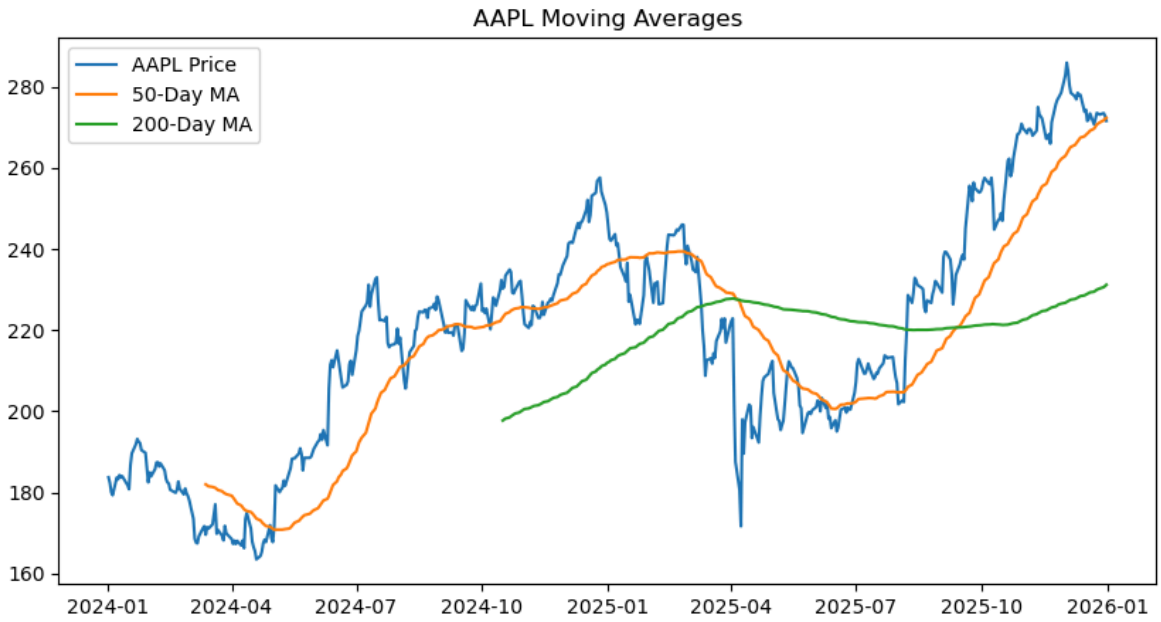
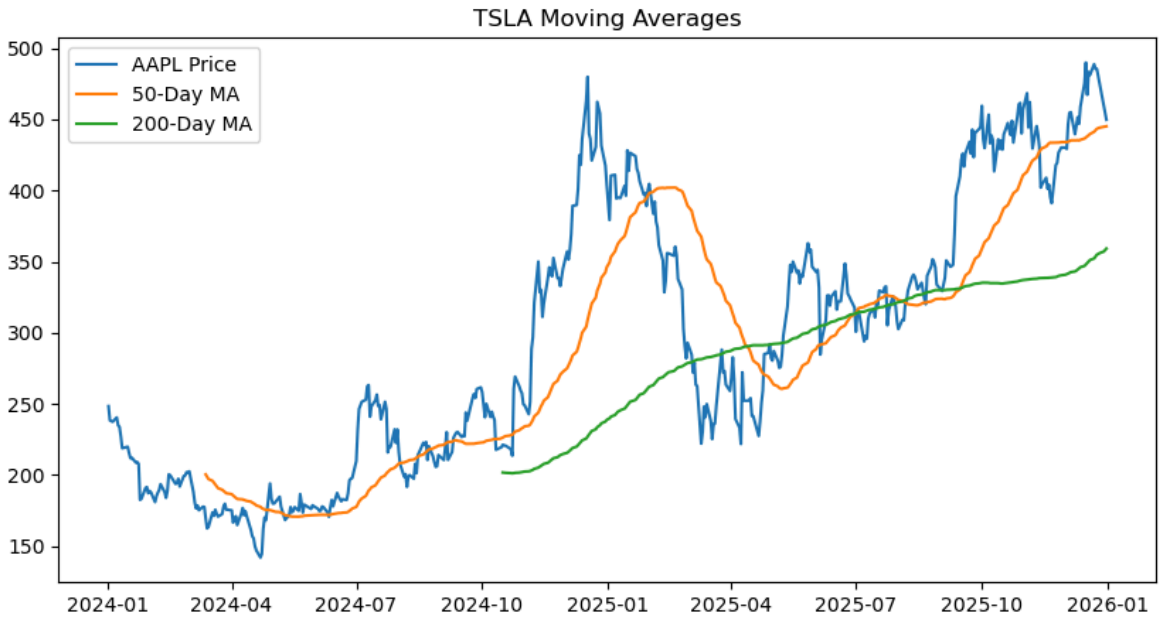
In [5]:
```
import yfinance as yf
import pandas as pd
import matplotlib.pyplot as plt

# Select stocks
stocks = ['AAPL', 'TSLA', 'MSFT', 'GOOGL']

# Download data (last 1 year)
data = yf.download(stocks, start='2024-01-01', end='2026-01-01',auto_adjust = Fa

# Show first few rows
print(data.head())

# 1. Calculate Daily Returns
returns = data.pct_change()


# 2. Calculate Risk & Return
mean_returns = returns.mean()
risk = returns.std()

print("\nAverage Returns:\n", mean_returns)
print("\nRisk (Volatility):\n", risk)

# 3. Plot Stock Prices
data.plot(figsize=(10,5))
plt.title("Stock Price Trends")
plt.show()

# 4. Moving Averages (AAPL example)
aapl = data['AAPL']
ma50 = aapl.rolling(window=50).mean()
ma200 = aapl.rolling(window=200).mean()

plt.figure(figsize=(10,5))
plt.plot(aapl, label='AAPL Price')
plt.plot(ma50, label='50-Day MA')
plt.plot(ma200, label='200-Day MA')
plt.legend()
plt.title("AAPL Moving Averages")
plt.show()

# 5. Moving Averages (TSLA example)
tsla = data['TSLA']
ma50_1 = tsla.rolling(window=50).mean()
ma200_1 = tsla.rolling(window=200).mean()

plt.figure(figsize=(10,5))
plt.plot(tsla, label='AAPL Price')
plt.plot(ma50_1, label='50-Day MA')
plt.plot(ma200_1, label='200-Day MA')
plt.legend()
plt.title("TSLA Moving Averages")
plt.show()
```

```
[*********************100%***********************]  4 of 4 completed
```

```
Ticker            AAPL       GOOGL        MSFT        TSLA
Date
2024-01-02  183.731323  137.037384  364.589447  248.419998
2024-01-03  182.355591  137.781250  364.324036  238.449997
2024-01-04  180.039658  135.271973  361.709106  237.929993
2024-01-05  179.317169  134.617386  361.522339  237.490005
2024-01-08  183.652100  137.701889  368.344757  240.449997

Average Returns:
 Ticker
AAPL     0.000933
GOOGL    0.001829
MSFT     0.000657
TSLA     0.001973
dtype: float64

Risk (Volatility):
 Ticker
AAPL     0.017571
GOOGL    0.019062
MSFT     0.013975
TSLA     0.040003
dtype: float64
```







Conclusions

Tech stocks showed consistent upward trends over the period
Tesla exhibited higher volatility compared to Microsoft and Apple
Moving averages indicated long-term bullish trends for AAPL

In [ ]: